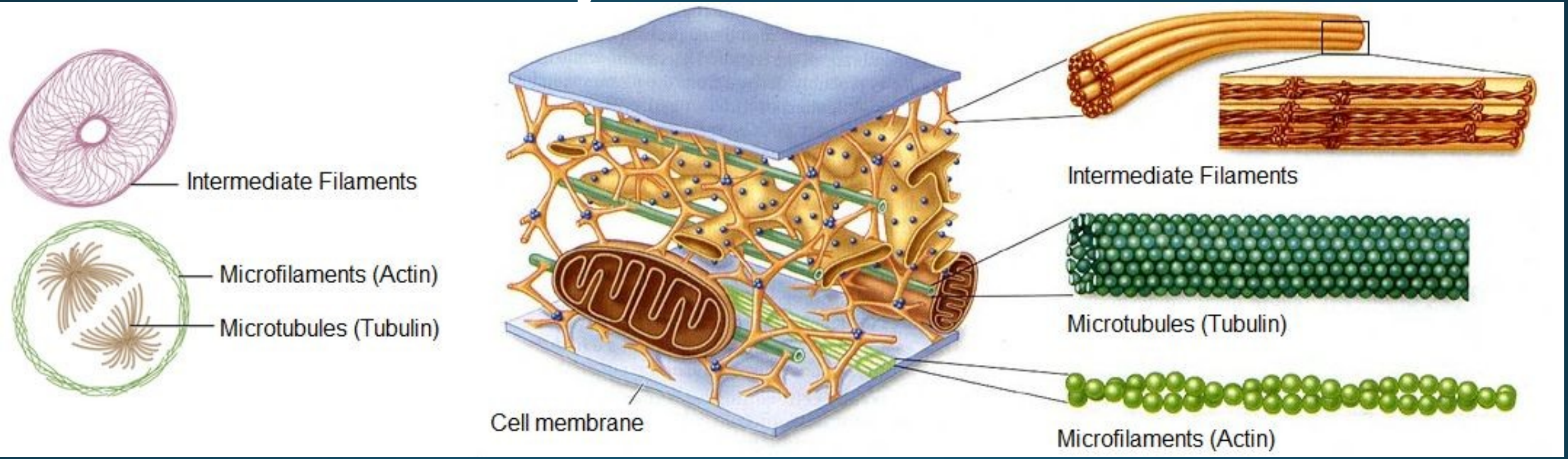
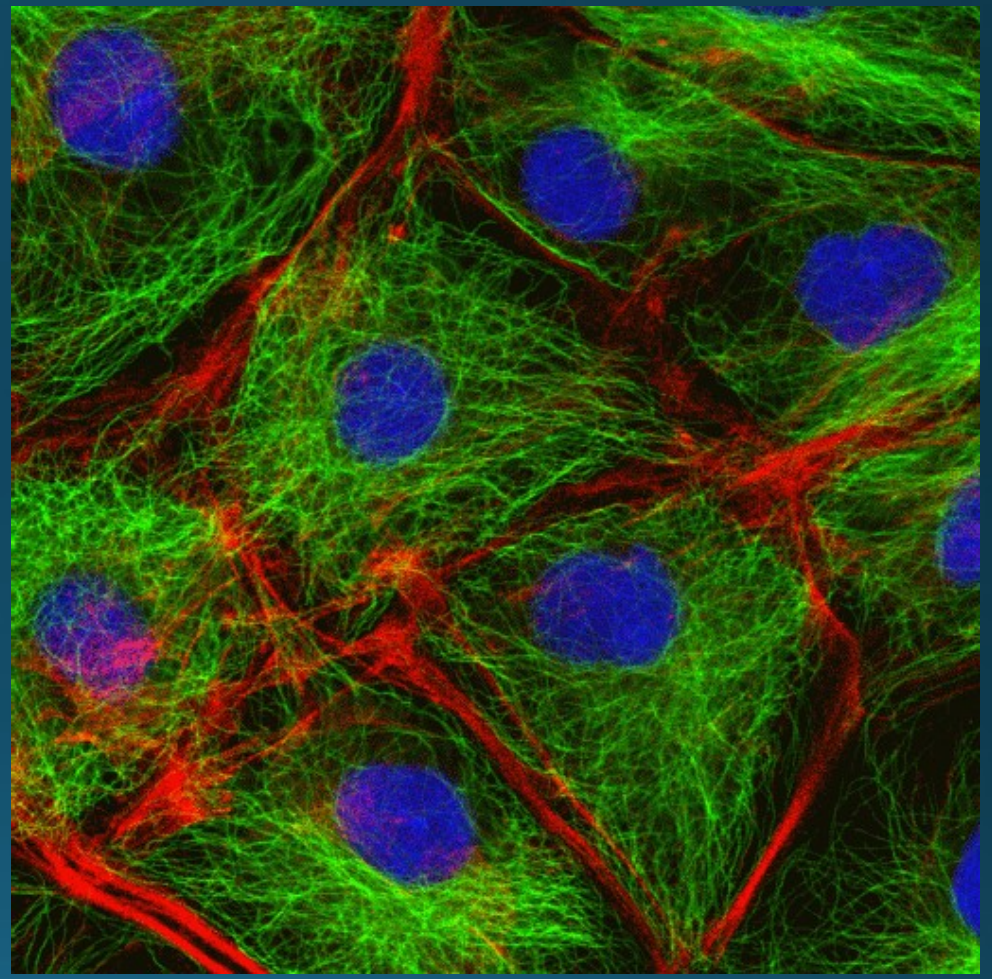
VII. Cytoskeleton

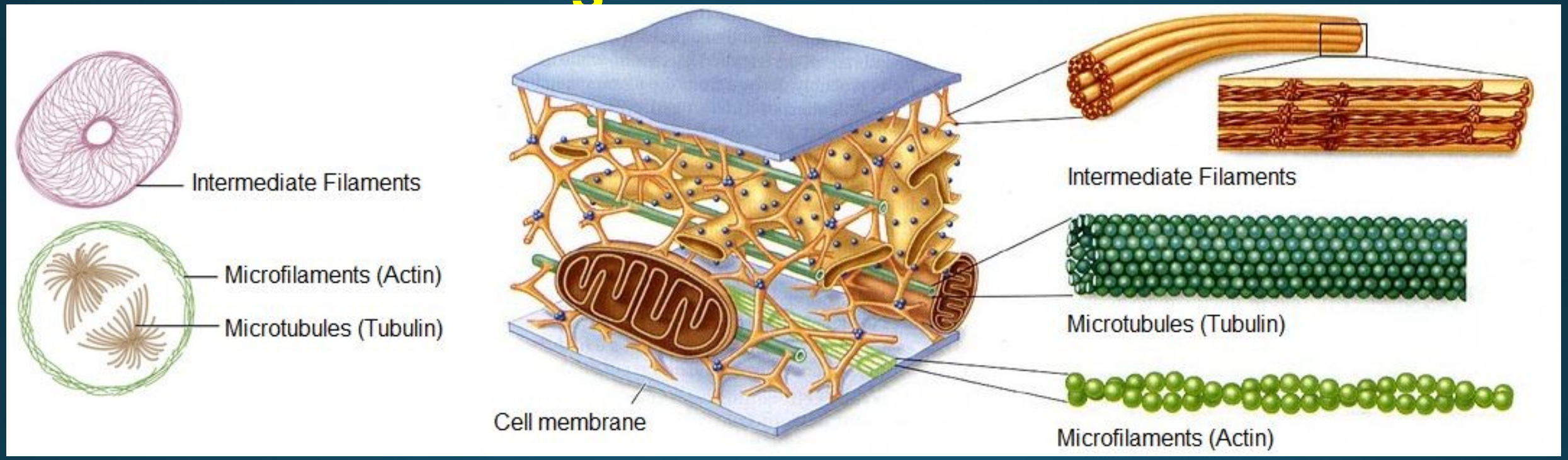


Network of connected filaments and tubules. It extends from the nucleus to the plasma membrane. EM showed the presence of an organized cytoplasm and protein fibers.

VII. Cytoskeleton

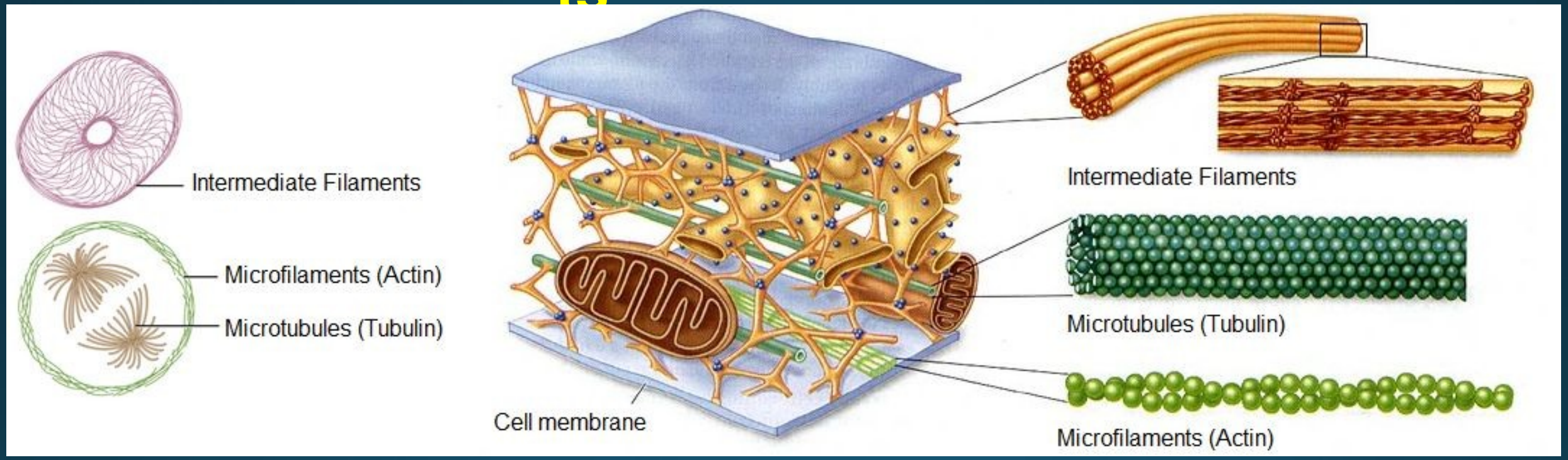


Microtubules



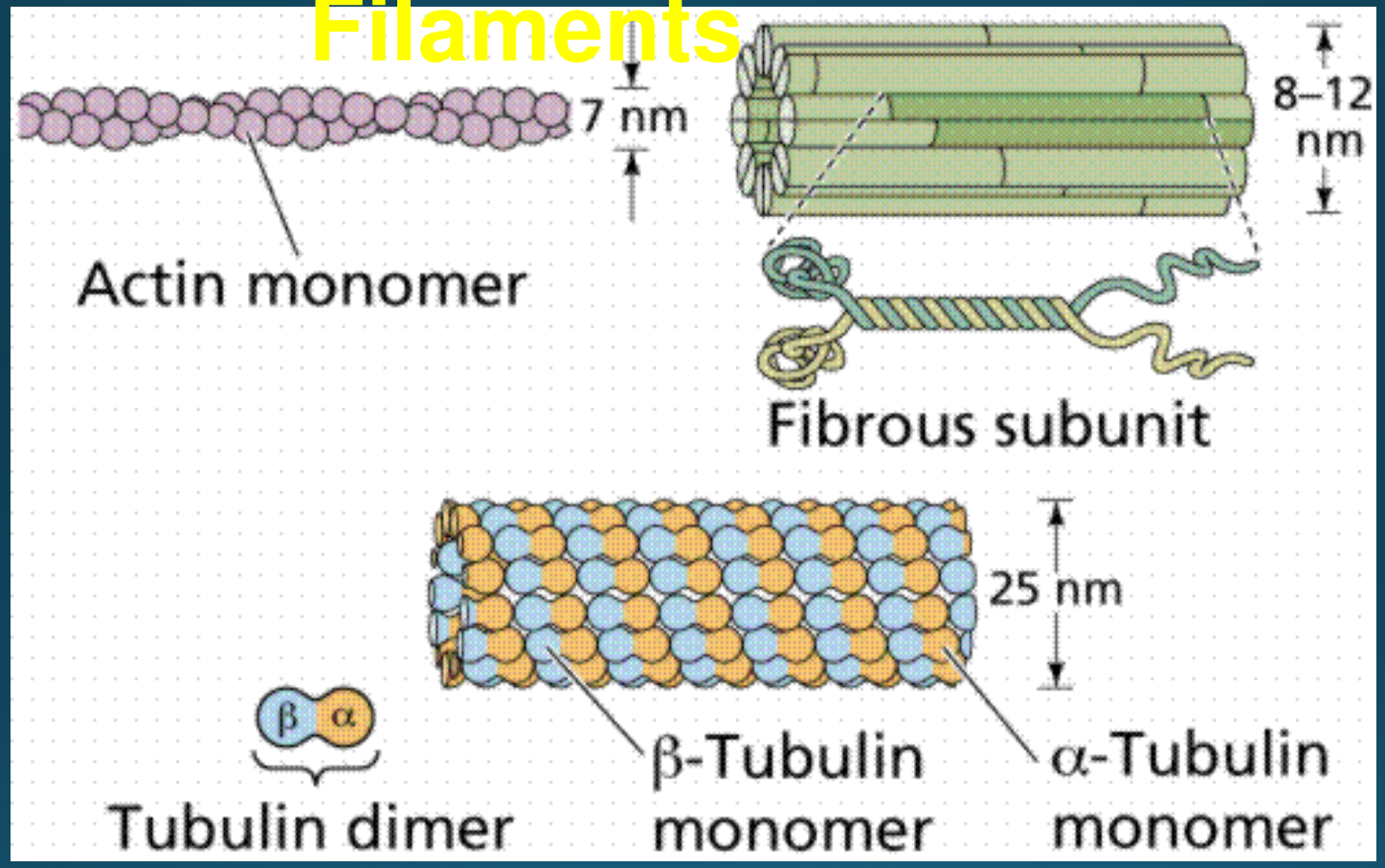
Small hollow cylinders composed of a globular protein tubulin.

Microfilaments



Similar to microtubules but it is a solid structure. Microtubules and Microfilaments serve as cytoskeleton supporting the cell to keep its size and shape.

Actin Filaments

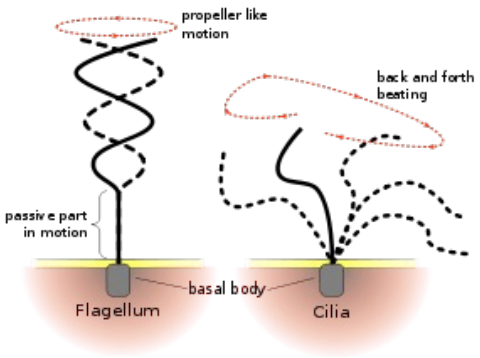


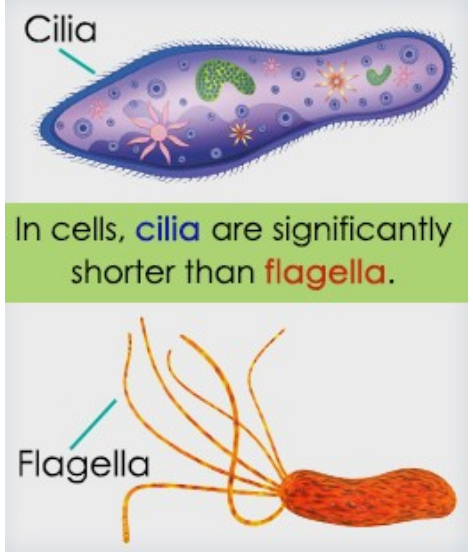
Composed of 2 chains of globular **Actin** monomers twisted to form a helix. Actin filaments form a dense complex web under the **plasma membrane**.

Intermediate filaments

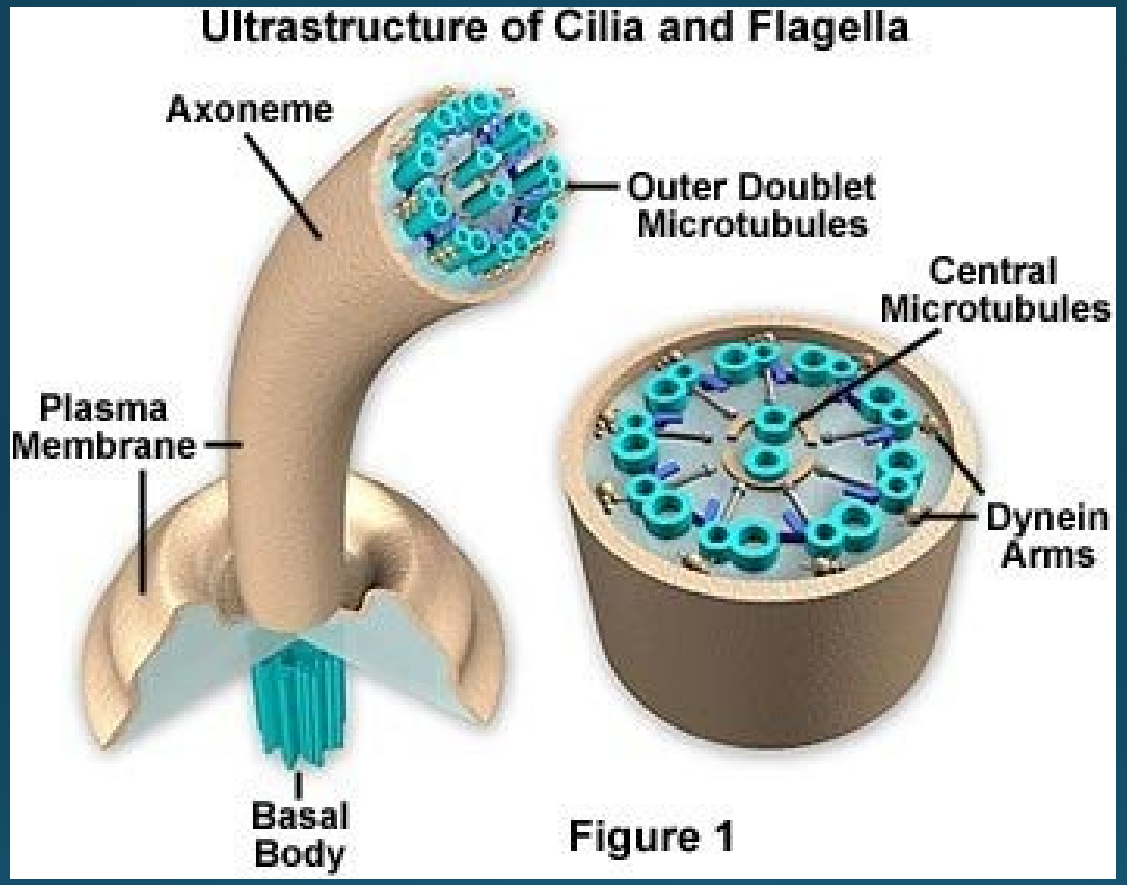
Cilia & Flagella

- Cilia are short, numerous, hair-like projection that can move in undulating fashion.
- Flagella are longer, fewer in number, projections that move in whip-like fashion.





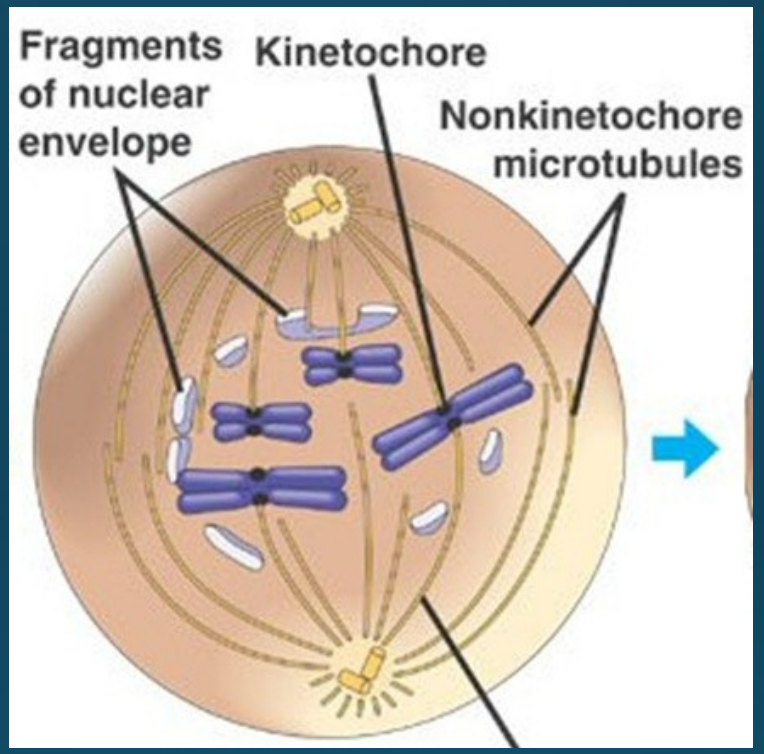
Cilia &

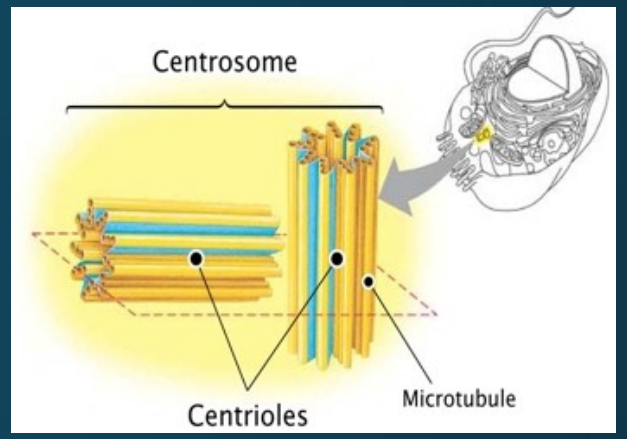


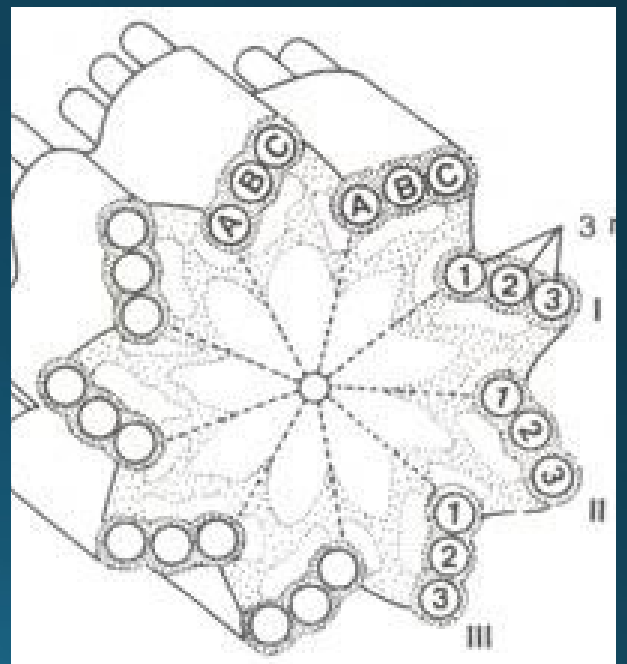
Cilia and Flagella have 9+2 arrangement of microtubules and both involved in cell movement.

Centrosome

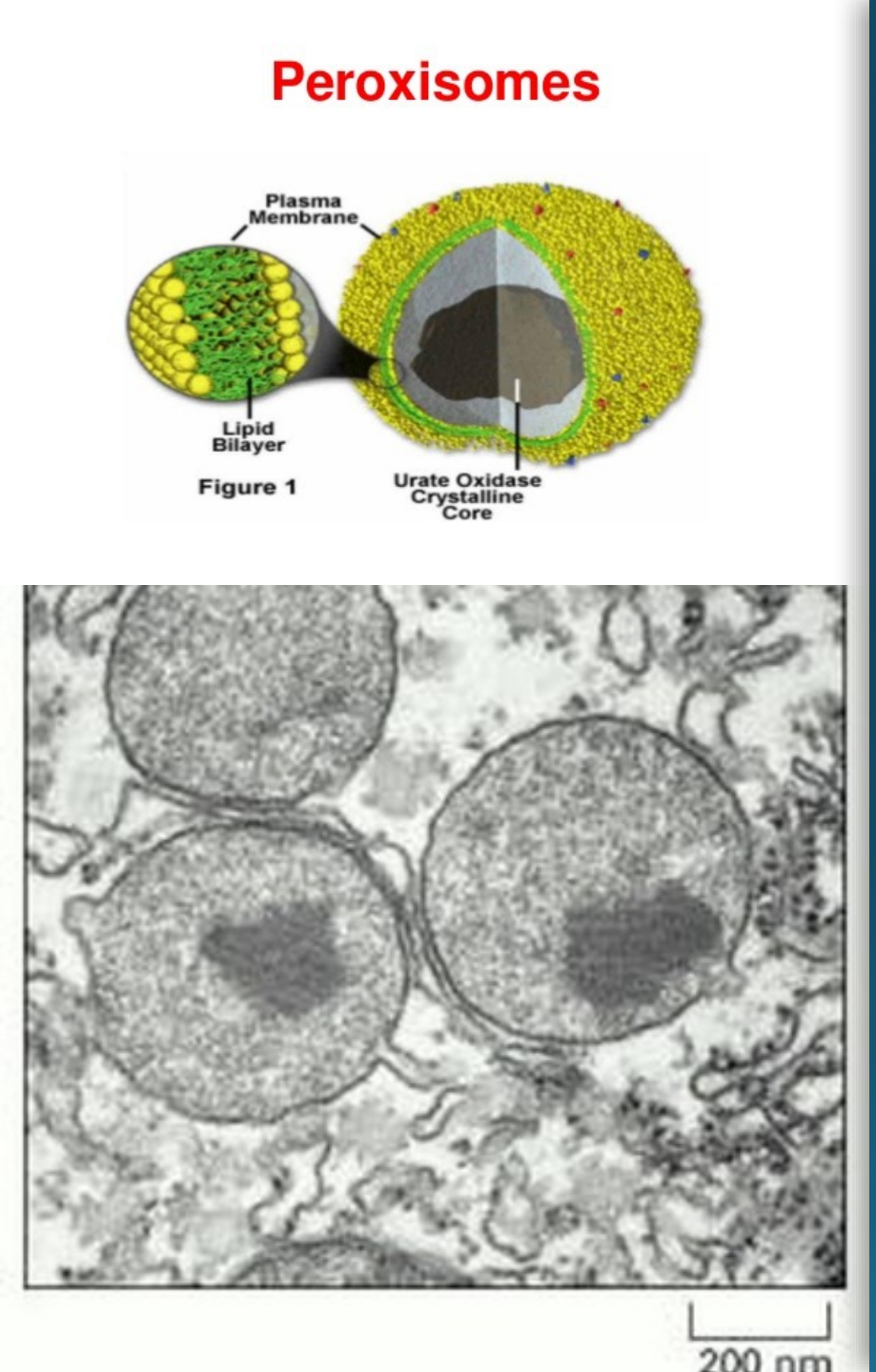
- Contains two centrioles lying at right angles to each other. Centrioles are short cylinders with 9+0 pattern of microtubules triplets.
- Centrioles are self replicating organelles which are often duplicated during prophase and metaphase.







3. Cross section of centriole showing nine set
bules.

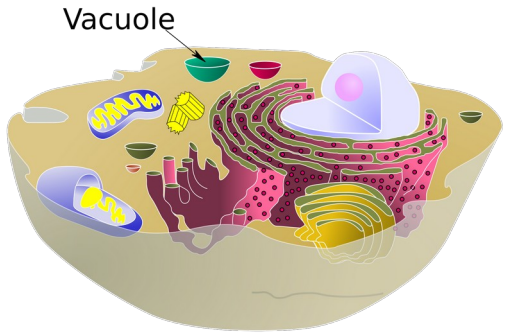


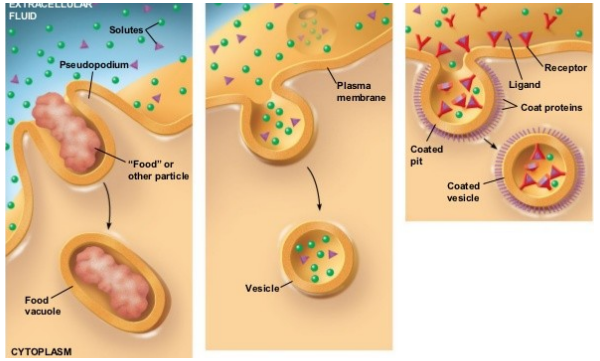
Peroxisomes

- Or Microbodies are common in liver cells and renal proximal tubules. They have a dense matrix enclosed by a single membrane and hold enzymes involved in beta and alpha oxidation of certain fatty acids and destruction of hydrogen peroxide H2O2.

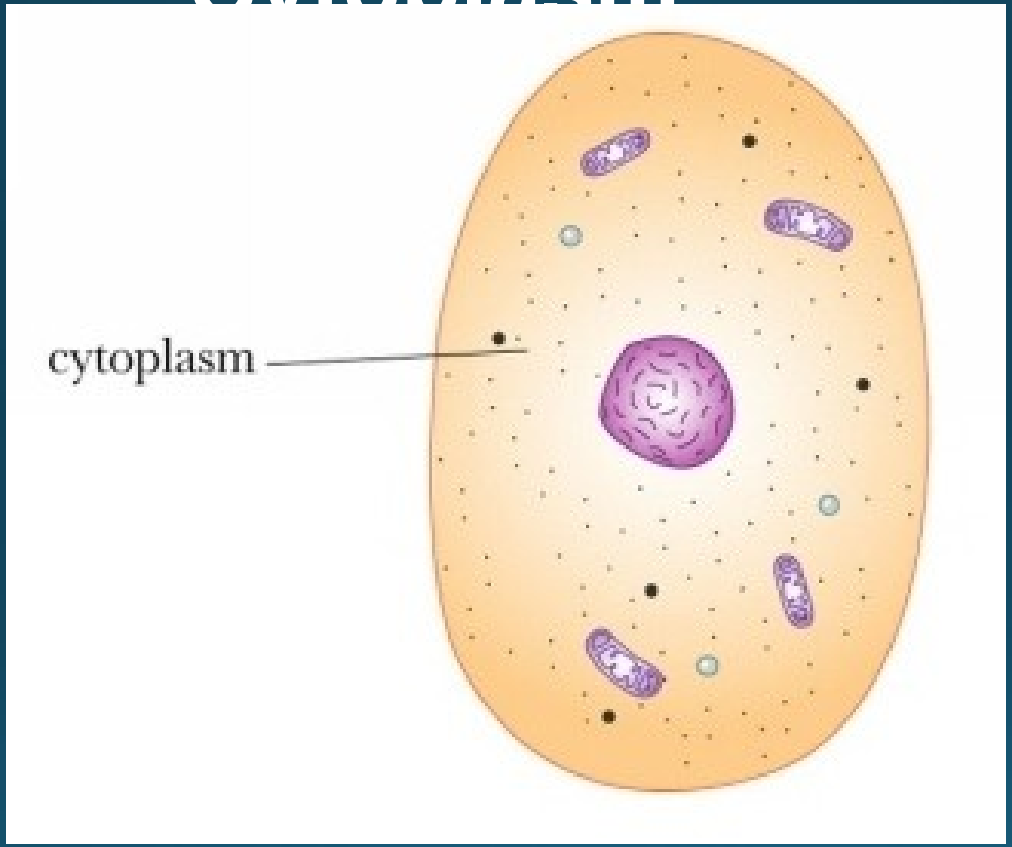
VIII. Vacuoles and Vesicles

- Vacuoles are single-membrane organelles that are essentially part of outside that are located within the cell (storage area).
- Vesicles are much smaller than vacuoles and function in transporting materials both within and outside the cell.

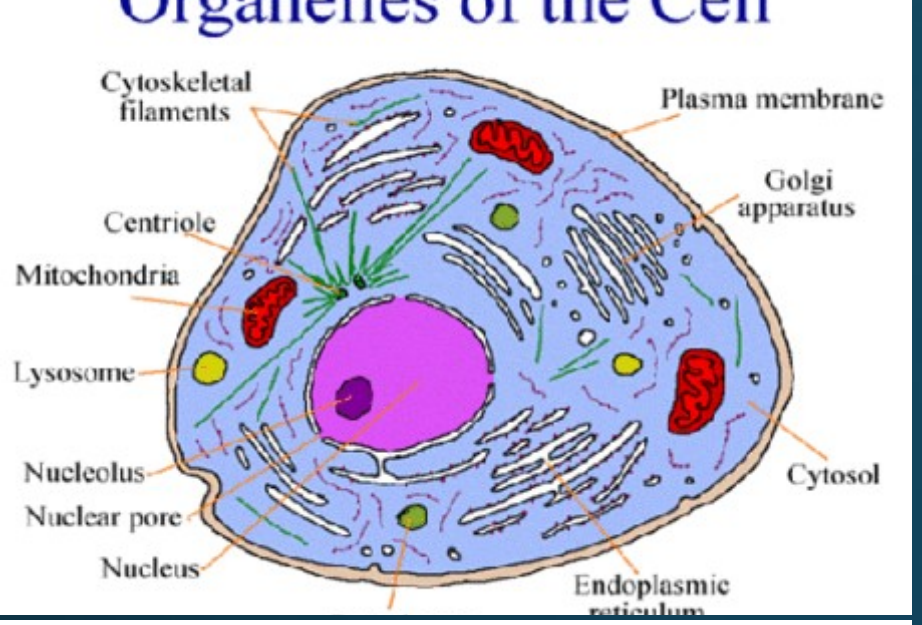




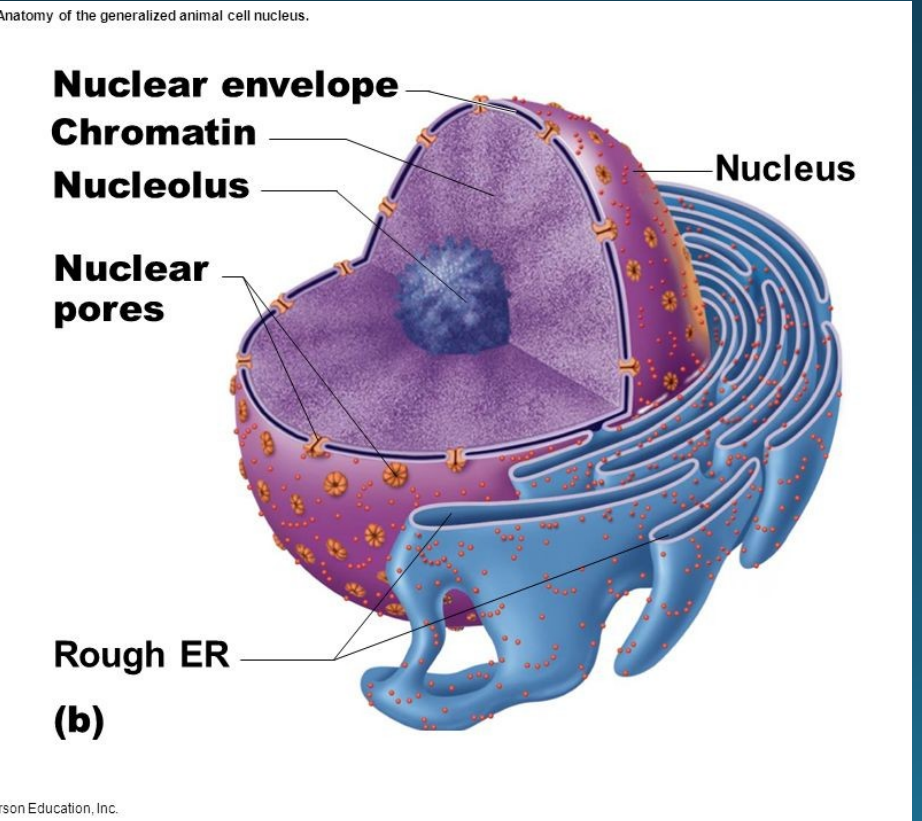
IX. Cytoplasm



- The material between the **plasma membrane** (cell membrane) and the **nuclear envelope**.



Anatomy of the generalized animal cell nucleus.



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X. Nucleus

- Double membrane organelle in eukaryotes contain nucleic acids.
- Numerous pores occur in the nuclear envelope allowing RNA and other chemicals to pass, but the DNA not to pass.
- Nucleus consists of 4 components: Nuclear membrane, nuclear sap, nucleolus and chromatin material.

Cellular Mechanisms

A. Cellular Respiration

Cells require energy for active transport, synthesis, impulse induction, contraction, and so on. Cells must be able to capture and store energy & release that energy in appropriate amounts when needed.

An important source of energy for cells is **glucose** (C6H12O6).

$$C_6H_{12}O_6 + O_2 \longrightarrow H_2O + CO_2 + \text{Energy} \uparrow$$

However, this reaction **releases huge amounts of energy**. So, cells gradually break down glucose in a whole series of reactions & use the smaller amounts of energy released in these reactions to produce ATP from ADP. Then, cells can break down ATP.

$$A\text{-}P^{+++}P^{+++}P \longleftrightarrow A\text{-}P^{+++}P + P^{\uparrow} + 7700 \text{ calories}$$

The energy released from this reaction is used by cells for active transport, synthesis, contraction and metabolism. Cells need large amounts of ATP and make more, but making ATP requires energy.

The break down of glucose releases energy, but **how the energy released in the breakdown of glucose used to make ATP?**

Oxidative Phosphorylation

A primary source of energy is **OXIDATION**. Specifically, cells use a type of oxidation called hydrogen transfer to generate energy.

The hydrogen transfer reaction so-named because pairs of hydrogen are transferred from one substrate to another.

In a cell, hydrogen transfer reactions occur in **MITOCHONDRIA**. Pairs of hydrogen are passed from one substrates to another and these substrates called **HYDROGEN CARRIERS**

These **hydrogen transfer reactions** release energy that is used to make ATP from ADP (in other words, to add a third phosphate to ADP in a reaction called **phosphorylation**).

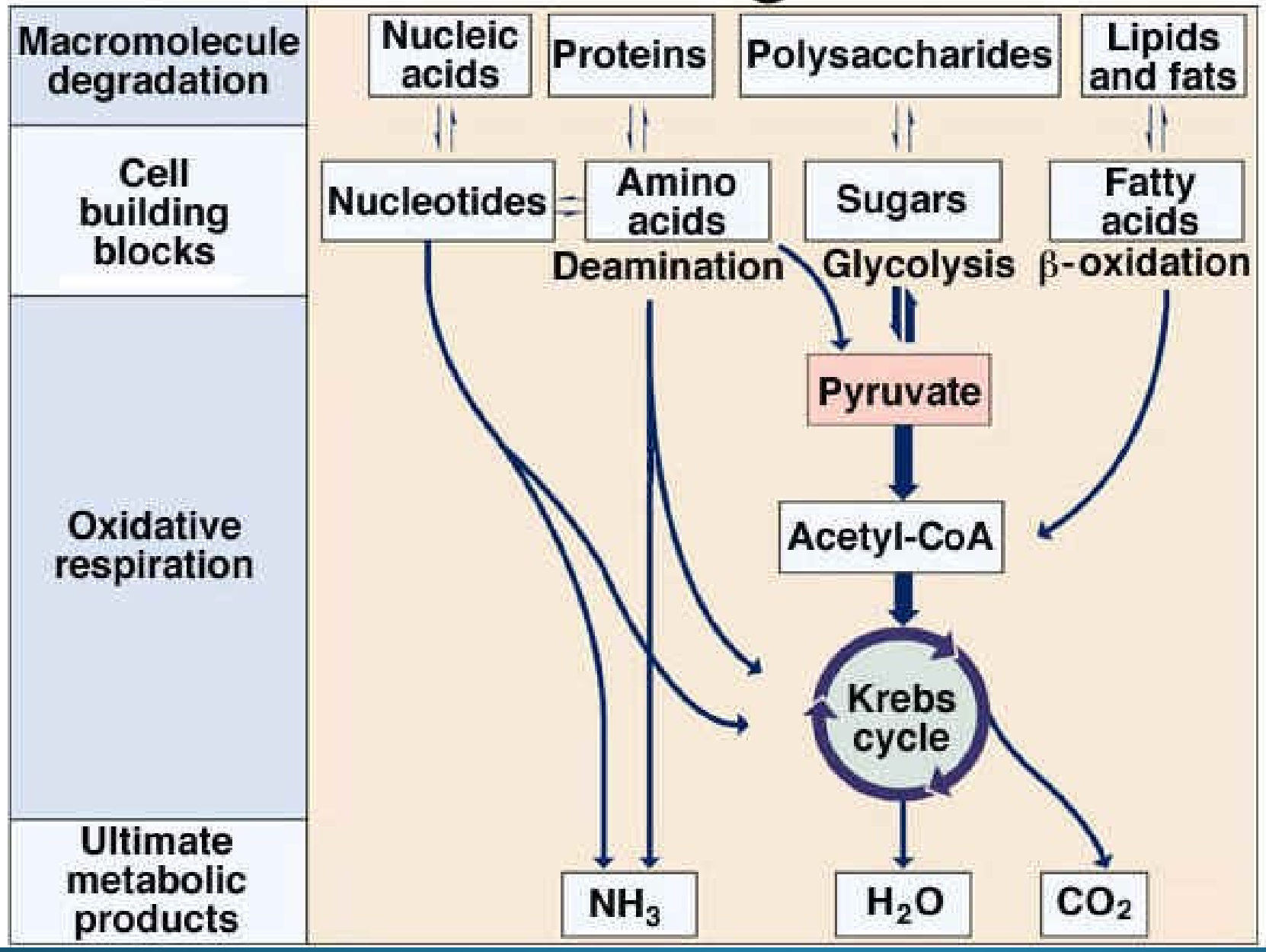
Oxidative phosphorylation produces lots of energy but requires hydrogen.

Where does the hydrogen come from?

Sources of hydrogen include **GLYCOLYSIS** and **KREB'S CYCLE**

GLYCOLYSIS: breakdown of glucose. Cells obtain glucose from the blood. Blood glucose maintained by two process : Glycogenesis and Glycogenolysis.

GLYCOGENESIS: the production of glycogen from glucose and occurs in the liver and skeletal muscles when blood levels are too high.

GLYCOGENOLYSIS: breakdown of glycogen to release individual molecules of glucose. This occurs when blood glucose levels begin to decline.

Catabolism of Other Organic Molecules